The Four Gateways: Setup Guide

Run the Flight Recorder demo and the four gateways labs on your own Mac or Windows laptop. DAMA Rocky Mountain, October 2026. Nathan Platter, Lincoln, NE.

SYNTHETIC SAMPLE DATA ONLY. Every name, number, and ID in this kit is invented. The labs run fully offline, and nothing leaves your laptop.

What you get

- Clickable demo: flight-recorder/index.html. Add ?gateways=1 to turn on the four gateways view.
- Python labs, one per gateway, in dama-2026/gateways/. They run in mock mode with no model installed.
- Optional local model through Ollama: llama3.1:8b or deepseek-r1:7b.
- agents/AGENTS.md and agents/guardrails.yaml: the same four gates written for an agent.

The four gateways at a glance

Gateway	Control	Lab	Demo step
G1. Use	Classify, redact, keep sensitive data on a local model	g1_classify_redact.py	Step 2
G2. Prompt	Locked instructions, injection defense, tool allow-list	g2_prompt_guard.py	Steps 1 and 5
G3. Context	Permissioned retrieval, minimum data, source trace	g3_context_scope.py	Steps 2 and 4
G4. Self-check and exit	Fail closed, named human, audit log	g4_selfcheck_exit.py	Step 3, Closer

1. Prerequisites

- Mac: macOS 14 Sonoma or newer for Ollama. Apple silicon (M1 or later) runs local models best.
- Windows: Windows 10 22H2 or newer, or Windows 11.
- 16 GB of RAM recommended for 7B or 8B models (8 GB works with smaller models such as llama3.2:3b).
- About 10 GB of free disk space for two models.
- A modern browser: Safari, Chrome, Edge, or Firefox.

- Python 3.10 or newer. The labs use only the standard library, so there is nothing to pip install.
- Git is optional. You can download a ZIP instead.

Check Python:

```
Mac:      python3 --version
Windows:  py --version
```

If Python is missing: on Mac, install it from python.org/downloads (or run xcode-select --install). On Windows, install it from python.org/downloads and tick "Add python.exe to PATH", or run:

```
winget install Python.Python.3.12
```

2. Install Ollama and pull the models (optional)

Skip this section if you only want mock mode. Mock mode shows every gateway with repeatable output.

Mac

- Download Ollama for macOS from ollama.com/download.
- Open the .dmg and drag Ollama into Applications. Start it once. It offers to add the ollama command to your PATH; accept.

Windows

- Download OllamaSetup.exe from ollama.com/download and run it. No administrator rights are needed.
- Ollama runs in the background, and the ollama command works in PowerShell or Command Prompt.

Pull the models (both platforms):

```
ollama pull llama3.1:8b      # about 4.9 GB
ollama pull deepseek-r1:7b   # about 4.7 GB
ollama list
```

Quick test: ollama run llama3.1:8b "Say hello in five words." The local API listens on http://localhost:11434.

3. Get the code

With Git:

```
git clone https://github.com/nathanplatteruser/gis-2027.git
cd gis-2027
```

Without Git: open github.com/nathanplatteruser/gis-2027, click Code, then Download ZIP. Unzip it and open a terminal in the unzipped folder.

4. Optional virtual environment

The labs need no packages, but a virtual environment keeps your experiments separate if you add any.

```
Mac:      python3 -m venv .venv && source .venv/bin/activate
Windows:  py -m venv .venv
          .venv\Scripts\Activate.ps1
```

If PowerShell blocks the activate script, run: Set-ExecutionPolicy -Scope CurrentUser RemoteSigned

5. Run in mock mode (no model)

```
cd dama-2026
Mac:      python3 run_all.py
Windows:  py run_all.py
```

You should see four sections and a summary ending in four PASS lines:

```
PASS G1 no sensitive value leaves
PASS G2 injected money move blocked
PASS G3 out-of-role doc denied
PASS G4 fails closed, writes nothing
```

Run the tests: python3 -m unittest -v test_gateways (Windows: py -m unittest -v test_gateways). Six tests should pass.

6. Run with a local model

```
Mac:
  FR_BACKEND=ollama FR_MODEL=llama3.1:8b python3 run_all.py
  FR_BACKEND=ollama FR_MODEL=deepseek-r1:7b python3 run_all.py

Windows PowerShell:
  $env:FR_BACKEND="ollama"; $env:FR_MODEL="llama3.1:8b"; py run_all.py
```

The model now writes the drafts. The gates still decide. A model may answer differently from the mock, so read each section: G2 blocks any tool not on the allow-list, G3 removes any sentence without a valid source, and G4 refuses to write a number when sources disagree. DeepSeek-R1 prints its reasoning; the labs strip it before checking. On a laptop without a GPU, expect 10 to 60 seconds per call.

7. Walk through each lab

G1. Use

Classifies each field of a synthetic applicant record, redacts the sensitive ones, and routes the raw record to local-only. Try it: add a new field with a phone number in free text and watch it get caught.

```
python3 gateways/g1_classify_redact.py
```

G2. Prompt

A broker note hides "ignore previous instructions" and asks for a $4,800 refund. The lab flags the injection and the code gate blocks move_money even if the model proposes it. Try it: add a tool to the allow-list and see what changes.

```
python3 gateways/g2_prompt_guard.py
```

G3. Context

Retrieves only documents the underwriting role may read (the HR file is denied), then removes any sentence whose number does not appear in its cited line. Try it: ask as the reinsurance role.

```
python3 gateways/g3_context_scope.py
```

G4. Self-check and exit

Three sources disagree on a treaty claim and one tries to instruct the AI. The lab discards the model's guess, writes no number, and sends one specific question to a named analyst. Every step lands in out/flight-recorder.jsonl.

```
python3 gateways/g4_selfcheck_exit.py
```

8. Open the clickable demo

- Open flight-recorder/index.html in your browser (double-click works), or use the live link: nathanplatteruser.github.io/gis-2027/flight-recorder/?gateways=1#step1
- Click "Four gateways" in the top bar to turn the overlay on or off.
- Step 2: use the raw vs redacted toggle. Step 3: the fail-closed record hands a question to a human. Closer: download the evidence pack.

9. Troubleshooting

- **"python3" or "py" not found:** Install Python (section 1). On Windows, reopen the terminal after install so PATH updates.
- **"Could not reach Ollama at http://localhost:11434":** Start the Ollama app, or run ollama serve in another terminal. Or unset FR_BACKEND to go back to mock mode.
- **"model not found":** Run ollama pull with the exact name in FR_MODEL, then ollama list.

- **Very slow or out of memory:** Close other apps, or use a smaller model: ollama pull llama3.2:3b and set FR_MODEL=llama3.2:3b.
- **Port 11434 already in use:** Ollama is already running. That is fine.
- **PowerShell will not activate the venv:** Set-ExecutionPolicy -Scope CurrentUser RemoteSigned, or skip the venv; it is optional.
- **Demo page looks blank:** Use a current browser. The demo is one HTML file with no downloads.
- **Refusing to send data off this machine:** OLLAMA_HOST points somewhere other than localhost. The labs only talk to a local model by design.
- **Test the Ollama path without downloading a model:** python3 tests/test_ollama.py starts a small stand-in for the Ollama API and runs every lab as llama3.1:8b and deepseek-r1:7b, including a "bad" model that obeys the injection, invents numbers, and echoes an SSN. All gates should still pass. Add --real llama3.1:8b to also run against your real Ollama. With make installed: make test-ollama.
- **A small model gives odd answers:** Expected. The gates decide the outcome, not the model. A model that forgets to cite sources gets its sentences removed by G3, which is the gate working.

10. Data handling

- Every record in this kit is synthetic and labeled SYNTHETIC SAMPLE DATA.
- Mock mode makes no network calls. Ollama mode calls only localhost; the code refuses any other host.
- The demo runs in your browser and uploads nothing. Intake answers are not saved.
- The audit log out/flight-recorder.jsonl stays in your folder. Delete it any time.
- Do not load real customer data into a teaching kit. Use it to design your own controls, then build them into your governed systems.

11. Use it in your own product

- Rewrite agents/AGENTS.md and agents/guardrails.yaml for your agent.
- Port the four checks: classify and redact, gate, retrieve with trace, reconcile and record.
- Keep one test per rule so the policy and the code cannot drift apart.

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